

Jacob Mattie

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Full-Stack Engineer focused on ETL systems, data infrastructure, and automation in production environments. Builds resilient pipelines and end-to-end platforms, integrating LLM-based Agents with strong context management to accelerate development and improve system reliability.

Python, C++, SQL, Bash | Django, Selenium | Linux (server ops)

BSc. Mathematics, 2022, Simon Fraser University

Work Experience

Investment Analyst

Aligned Capital Partners / River Wealth

Vancouver, BC

11/2024 - Ongoing

Sole developer and system administrator responsible for greenfield development, including Linux server provisioning and data infrastructure design, supporting a Django platform integrating client and market data with regulated internal records.

- Engineered distributed scraping pipelines using Selenium and Celery workers, capturing and tracking 35,000+ daily financial data points with **stateful retries**, **validation**, and fault-tolerant logging.
- Structured the analytics platform to support future ML workflows, allowing progression from heuristic strategies to **quantitatively validated** investment models as data volume grows.
- Introduced **LLM-assisted** development workflows, leveraging Agents with structured context windows to accelerate iteration, automate boilerplate generation, and improve debugging efficiency across the codebase.
- Reduced aggregate operational workload by 20% through targeted Python **automation** and process standardization.

Co-Founder

Liquid Energy

Calgary, AB

02/2024 - 08/2024

Test Technologist

IBC Technologies

Burnaby, BC

6/2022 - 9/2023

Hydronics lab technologist validating mechanical and digital performance of boilers, performing data analysis and infrastructure development, pioneering IoT integration and automation.

- Automated hardware **validation processes** by leveraging IoT microcontrollers and PLC systems, streaming test data into cloud-connected repositories for asynchronous analysis.
- Built high-throughput **data analysis** workflows in MATLAB for multi-million-point datasets, reducing certification validation time by over 600%.

Section Editor, Writer

The Peak Publications Society

Burnaby, BC

10/2020 - 6/2022

IT Technician (Internship)

BC Cancer Research Centre (BCCRC)

Vancouver, BC

09/2018 - 01/2019

Level 1-2 IT Support Technician at a scientific research institute providing end-user support, network administration assistance, and system configuration across laboratory and administrative environments.

- Managed user access controls and permissions within a segregated research network, enforcing data isolation and cybersecurity policies across multiple user groups.
 - Assisted in network recovery during facility-wide outage caused by a malware incident, supporting administrators in restoring connectivity and validating system integrity across endpoints.
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Volunteering

Automation Technologist

TRIUMF - Canada's Particle Accelerator

Vancouver, British Columbia

05/2025 - Ongoing

- Contributed the technical foundations that led to 3 scientific manuscripts in preparation on laboratory control systems and automation workflows.
- Developed a modular control-system deployment pipeline using Bash and containerization, standardizing environment provisioning for students, researchers, and faculty across heterogeneous computing environments.
- Engineered a Raspberry Pi-based stepper motor control system, including hardware wiring, GPIO-level driver implementation, and Python-based control interface through a transistor array for reliable, repeatable experimental control.